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# Paper Code : 21308

## F–408

## B. C. A. (Second Semester) EXAMINATION, 2018

### (New Course)

Paper No. BCA—N–203

MATHEMATICS—II

*Time : Three Hours ]* *[ Maximum Marks : 70*

**Note :** Attempt any *five* questions. All questions carry equal marks.

1. (a) Find the limiting points of the set :

$$\left\{1, -1, 1\frac{1}{2}, -1\frac{1}{2}, 1\frac{1}{3}, -1\frac{1}{3}, \ldots\right\}$$

(b) State and prove Bolzano-Weierstrass theorem.

2. (a) Evaluate $\lim_{x \to 0} \dfrac{(1+x)^{1/3} - (1-x)^{1/3}}{x}$.

(b) Show that a subset A of R is closed iff $\bar{A} = A$.

3. (a) Examine the continuity of the function at $x = 0$.

$f(x) = \dfrac{e^{x^{1/x}}}{1 + e^{1/x}}$, when $x \neq 0$, $f(0) = 0$.



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(b) Show that the sequence $\left\langle \dfrac{n}{n+1} \right\rangle$ is bounded $\forall\, n \in \mathrm{N}$.

4. (a) If $\langle x_n \rangle$ is a sequence such that $x_n > 0\ \forall\, n \in \mathrm{N}$ and $\lim_{n \to \infty} x_n = a$, prove that :

$$\lim_{n \to \infty} (x_1 x_2 x_3 \ldots\ldots x_n)^{1/n} = a$$

(b) Prove that the sequence $\langle x_n \rangle$, where $x_n = \sqrt{n+1} - \sqrt{n}$ converges and find its limit.

5. (a) Test the convergence of the series whose $n^{\text{th}}$ term is $\left(\dfrac{1}{n} - \log \dfrac{n+1}{n}\right)$.

(b) Test the series :

$$1 + \frac{3}{2}x + \frac{5}{9}x^2 + \frac{7}{28}x^3 + \cdots\cdots + \frac{2n+1}{n^3+1}x^n + \ldots$$

6. Test the convergence of the series

$$1 + \frac{3}{7}x + \frac{3.6}{7.10}x^2 + \frac{3.6.9}{7.10.13}x^3 + \frac{3.6.9.12}{7.10.13.16}x^4 + \ldots$$

7. (a) State and prove Lagrange's mean value theorem.

(b) Find the expansion of $\log(1+x)$ by using Maclaurin's theorem.

8. (a) Discuss the applicability of Rolle's theorem in the interval $[-1, 1]$ to the function $f(x) = |x|$.

(b) Find the values of $a$ and $b$ in order that

$$\lim_{x \to 0} \frac{x(1 + a\cos x) - b\sin x}{x^3} = 1.$$



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